

SQL Exercise: CRUD, Constraints, and Queries

Part 1: Create Tables with Constraints

Create the following tables with appropriate data types and constraints (including primary keys (PK), foreign keys (FK), and checks):

- Customers Table: Store customer information.
- Orders Table: Store order information for customers.
- Products Table: Store product details.
- OrderDetails Table: Store details about each product in the order (many-to-many relationship).

1. Create the Customers Table

Table Name: Customers

Column Name	Data Type	Constraint
customer_id	Integer	PK
first_name	Text	Not Empty
last_name	Text	Not Empty
email	Varchar(50)	No duplicates
created_at	Datetime	

2. Create the Products Table

Table Name: Products

Column Name	Data Type	Constraint
product_id	Integer	PK, autoincrement
product_name	Text	Not Empty
price	Real	Not Empty
stock_quantity	Integer	>0 , not null

3. Create the Orders Table

Table Name: Products

Column Name	Data Type	Constraint
order_id	Integer	PK, autoincrement

customer_id	Integer	FK to customer_id in customers
order_date	Datetime	current
Total_amount	real	>0 , not null

4. Create the OrderDetails Table

“ This is made for you, just copy and paste the code”

```
CREATE TABLE OrderDetails (
    order_detail_id INTEGER PRIMARY KEY AUTOINCREMENT, -- PK: unique identifier for
each order detail, Integer type
    order_id INTEGER, -- Foreign Key reference to Orders
    product_id INTEGER, -- Foreign Key reference to Products
    quantity INTEGER CHECK(quantity > 0), -- Check constraint ensures quantity is positive,
INTEGER type
    price REAL NOT NULL, -- Price of the product at the time of order, REAL type
    FOREIGN KEY (order_id) REFERENCES Orders(order_id), -- FK: establishes relationship
with Orders table
    FOREIGN KEY (product_id) REFERENCES Products(product_id) -- FK: establishes
relationship with Products table
);
```

Part 2: Insert Sample Data

Now, insert the following sample data into the tables:

1. Insert Customers

```
INSERT INTO Customers (first_name, last_name, email, phone_number)
VALUES ('John', 'Doe', 'john.doe@example.com', '1234567890'),
```

```
('Jane', 'Smith', 'jane.smith@example.com', '0987654321');
```

2. Insert Products

```
INSERT INTO Products (product_name, price, stock_quantity)
VALUES ('Laptop', 1000, 10),
      ('Phone', 500, 20),
      ('Tablet', 300, 15);
```

3. Insert Orders

```
INSERT INTO Orders (customer_id, total_amount)
VALUES (1, 1500),
      (2, 800);
```

4. Insert OrderDetails

```
INSERT INTO OrderDetails (order_id, product_id, quantity, price)
VALUES (1, 1, 1, 1000),
      (1, 2, 1, 500),
      (2, 3, 2, 300);
```

Part 3: Practice Queries

1. Retrieve all customers with their email addresses:
2. Retrieve all orders with a total amount greater than 1000:
3. `SELECT * FROM Products ORDER BY price DESC;`
4. Retrieve all orders along with customer information.
5. Retrieve the total sales (quantity * price) for each product. Only show products with 6 total sales greater than 1000:
6. Retrieve customers whose last name starts with 'S':
7. `SELECT * FROM Customers WHERE last_name LIKE 'S%';`
8. retrieve the total number of orders placed by each customer?
9. Which query retrieves all customers who have ordered a product with a price greater than 500?
10. Which query retrieves products that have been ordered more than once, including product details?